EDA Report - Marketing Campaigns Performance

As a marketing agency, our primary objective is to maximize the return on investment (ROI) for our clients' advertising campaigns. We have conducted two ad campaigns, one on Facebook and the other on AdWords, and we need to determine which platform yields better results in terms of clicks, conversions, and overall cost-effectiveness. By identifying the most effective platform, we can allocate our resources more efficiently and optimize our advertising strategies to deliver better outcomes for our clients.

Most Effective Ad Platform in terms of conversions, clicks, and overall cost-effectiveness

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Data Description

The dataset comprises a collection of data comparing the performance of two separate ad campaigns conducted throughout the year 2019. Specifically, the data covers a Facebook Ad campaign and an AdWords Ad campaign.
For each day of the year 2019, there is a corresponding row in the dataset, resulting in a total of 365 lines of campaign data to analyze. The dataset includes various performance metrics for each ad campaign, providing insights into their effectiveness and efficiency over time.

Key features included in the dataset are as follows:
- Date: The date corresponding to each row of campaign data, ranging from January 1st, 2019, to December 31st, 2019.
- Ad Views: The number of times the ad was viewed.
- Ad Clicks: The number of clicks received on the ad.
- Ad Conversions: The number of conversions resulting from the ad.
- Cost per Ad: The cost associated with running the Facebook ad campaign.
- Click-Through Rate (CTR): The ratio of clicks to views, indicating the effectiveness of the ad in generating clicks.
- Conversion Rate: The ratio of conversions to clicks, reflecting the effectiveness of the ad in driving desired actions.
- Cost per Click (CPC): The average cost incurred per click on the ad.

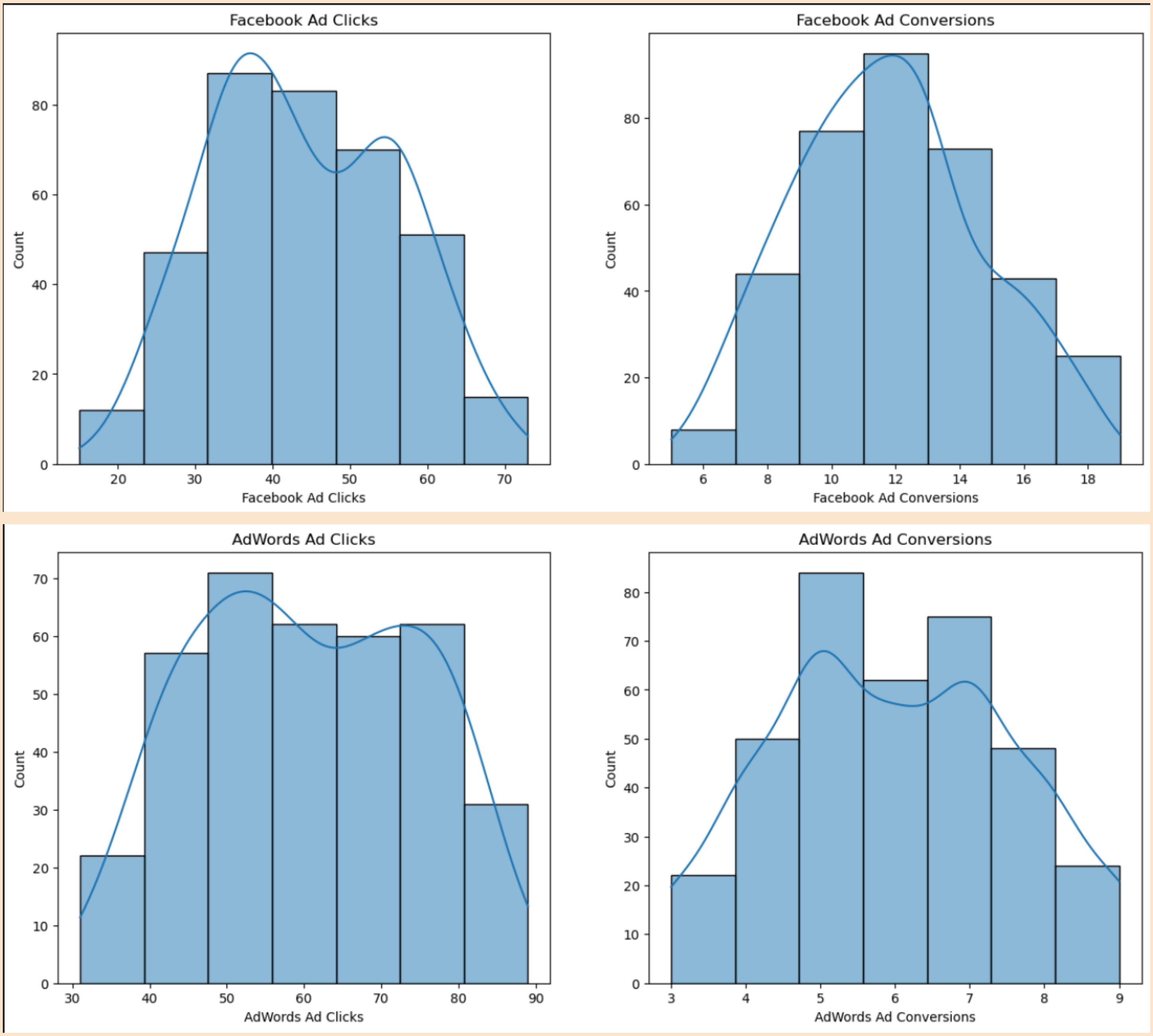
Data Overview

```python
df.head()
```

Python

	Date	Facebook Ad Campaign	Facebook Ad Views	Facebook Ad Clicks	Facebook Ad Conversions	Cost per Facebook Ad	Facebook Click-Through Rate (Clicks / View)	Facebook Conversion Rate (Conversions / Clicks)	Facebook Cost per Click (Ad Cost / Clicks)	AdWords Ad Campaign	AdWords Ad Views	AdWords Ad Clicks	AdWords Ad Conversions	Cost per AdWords Ad	AdWords Click-Through Rate (Clicks / View)	AdWords Conversion Rate (Conversions / Click)	AdWords Cost per Click (Ad Cost / Clicks)
0	1/1/2019	FB_Jan19	2116	18	8	$126	0.83%	42.73%	$7.14	AW_Jan19	4984	59	5	$194	1.18%	8.40%	$3.30
1	1/2/2019	FB_Jan19	3106	36	12	$104	1.15%	34.04%	$2.91	AW_Jan19	4022	71	6	$75	1.77%	7.80%	$1.05
2	1/3/2019	FB_Jan19	3105	26	8	$102	0.84%	31.45%	$3.89	AW_Jan19	3863	44	4	$141	1.13%	9.59%	$3.23
3	1/4/2019	FB_Jan19	1107	27	9	$71	2.45%	34.76%	$2.62	AW_Jan19	3911	49	5	$141	1.26%	11.08%	$2.86
4	1/5/2019	FB_Jan19	1317	15	7	$78	1.10%	47.59%	$5.38	AW_Jan19	4070	55	7	$133	1.36%	12.22%	$2.40

Comparing Campaigns performance



Metric	Platform Observation
Volume	AdWords drives significantly higher click volume (up to 90) compared to Facebook (up to 70).

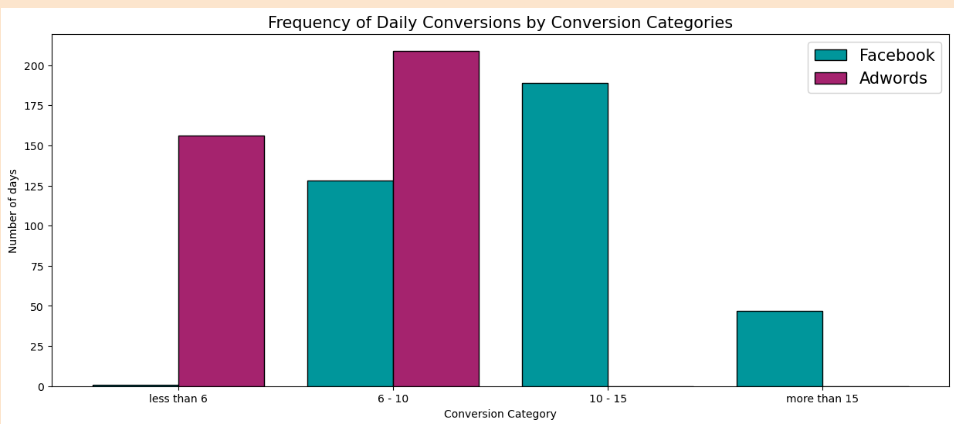
Efficiency	Facebook demonstrates higher conversion efficiency, achieving more conversions (up to 18) with lower clicks count compared to AdWords.
Stability	Facebook's unimodal curves indicate more **s**table and predictable performance metrics.

The distributions for both clicks and conversions across platforms exhibit a relatively symmetrical shape, suggesting that performance metrics are evenly distributed around the mean. This indicates a stable process with minimal outliers on either the high or low ends of the spectrum.

High & Low Conversion Days

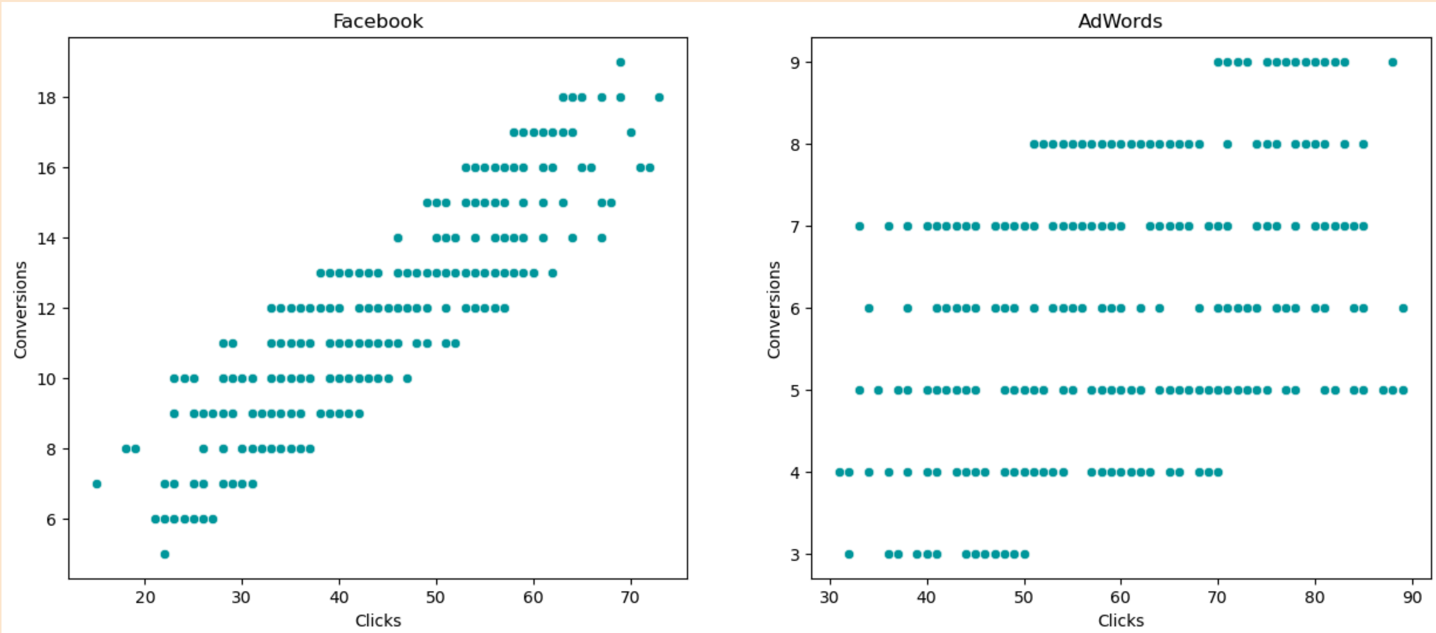
Count_x = Facebook Day Count
Count_y = AdWords Day Count

	Category	count_x	count_y
3	less than 6	1	156.0
1	6 - 10	128	209.0
0	10 - 15	189	0.0
2	more than 15	47	0.0



- The data suggests Facebook had higher frequent conversion days than AdWords, which either had very low conversion rates (less than 6) or moderate ones (6 - 10).
- There is a significant variance in the number of high-conversion days between two different campaigns.
- The absence of any days with conversions between 10 - 15 and more than 15 in AdWords indicates a need to review what strategies were changed or what external factors could have influenced these numbers.

Clicks vs Conversions



1. Facebook: Strong Positive Linear Relationship

- Insight: Displays a highly linear trend where conversions scale predictably with click volume. The tight clustering suggests that Facebook traffic is consistently qualified and intent-driven.
- Impact: High efficiency. Increasing clicks is a direct and reliable driver of conversion growth.

2. AdWords: Moderate & Dispersed Relationship

- Insight: Shows a widely dispersed (noisy) pattern. Similar click counts result in a broad range of outcomes (e.g., 30–50 clicks can yield anywhere from 3 to 7 conversions).

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Correlation Coeff
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Facebook : 0.87
AdWords :  0.45
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- **Facebook (High Impact):** A strong correlation ($r = 0.87$) confirms Facebook clicks are a primary driver of conversion rates. Scaling investment here offers the most direct path to growth.
- **Facebook Efficiency:** Because Facebook has high efficiency engagement which explains most conversion variance, prioritizing this channel ensures high-predictability returns.
- **AdWords (Moderate Impact):** AdWords shows a moderate correlation ($r = 0.45$) with **conversions**. It contributes to results, but the relationship is less direct than Facebook.
- **AdWords Optimization:** The 0.45 coefficient suggests external factors influence AdWords **conversion rates**. Further analysis is required to identify and mitigate these variables for better ROI.

Hypothesis Testing

Hypothesis: Advertising on Facebook will result in a greater number of conversions compared to advertising on AdWords.

Null Hypothesis (H0): There is no difference in the number of conversions between Facebook and AdWords, or the number of conversions from AdWords is greater than or equal to those from Facebook.

H0: $\mu_{Facebook} \leq \mu_{AdWords}$

Alternate Hypothesis (H1): The number of conversions from Facebook is greater than the number of conversions from AdWords.

H1: $\mu_{Facebook} > \mu_{AdWords}$

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Mean Conversion
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Facebook : 11.74
AdWords : 5.98

T statistic 32.88402060758184
p-value 9.348918164530465e-134

p-value is less than significance value, Reject the null hypothesis
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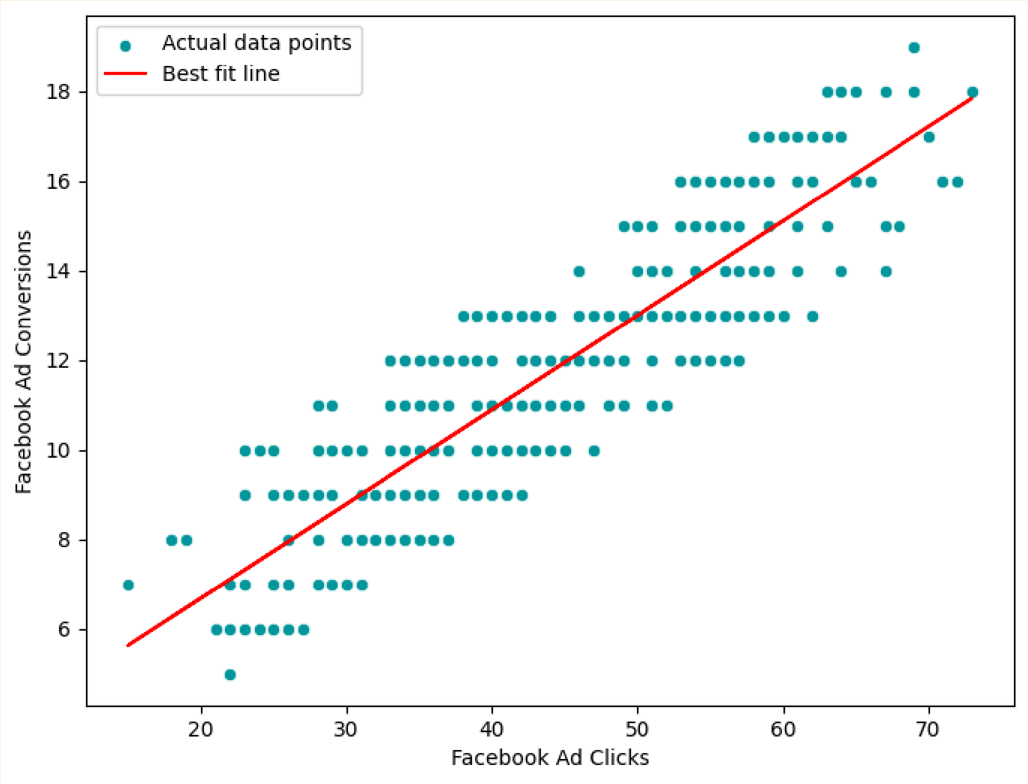
- **Superior Facebook Performance:** Facebook significantly outperforms AdWords with a mean conversion of **11.74** versus **5.98**, proving it is the more effective channel for generating results.
- **Strong Statistical Variance:** A massive T-statistic (**32.88**) confirms a substantial, distinct gap between the performance of the two platforms.
- **Highly Significant Results:** An extremely low p-value (9.35×10^{-134}) provides overwhelming evidence to reject the null hypothesis, confirming the findings are not due to random chance.
- **Validated Hypothesis:** Data strongly supports the alternate hypothesis: Facebook's conversion volume is definitively greater than AdWords'.
- **Strategic Reallocation:** Based on this statistical certainty, resources should be shifted toward Facebook—specifically by increasing spend and expanding ad formats—to maximize total conversions.

Regression Analysis

`How many facebook ad conversions can I expect given a certain number of facebook ad clicks?`

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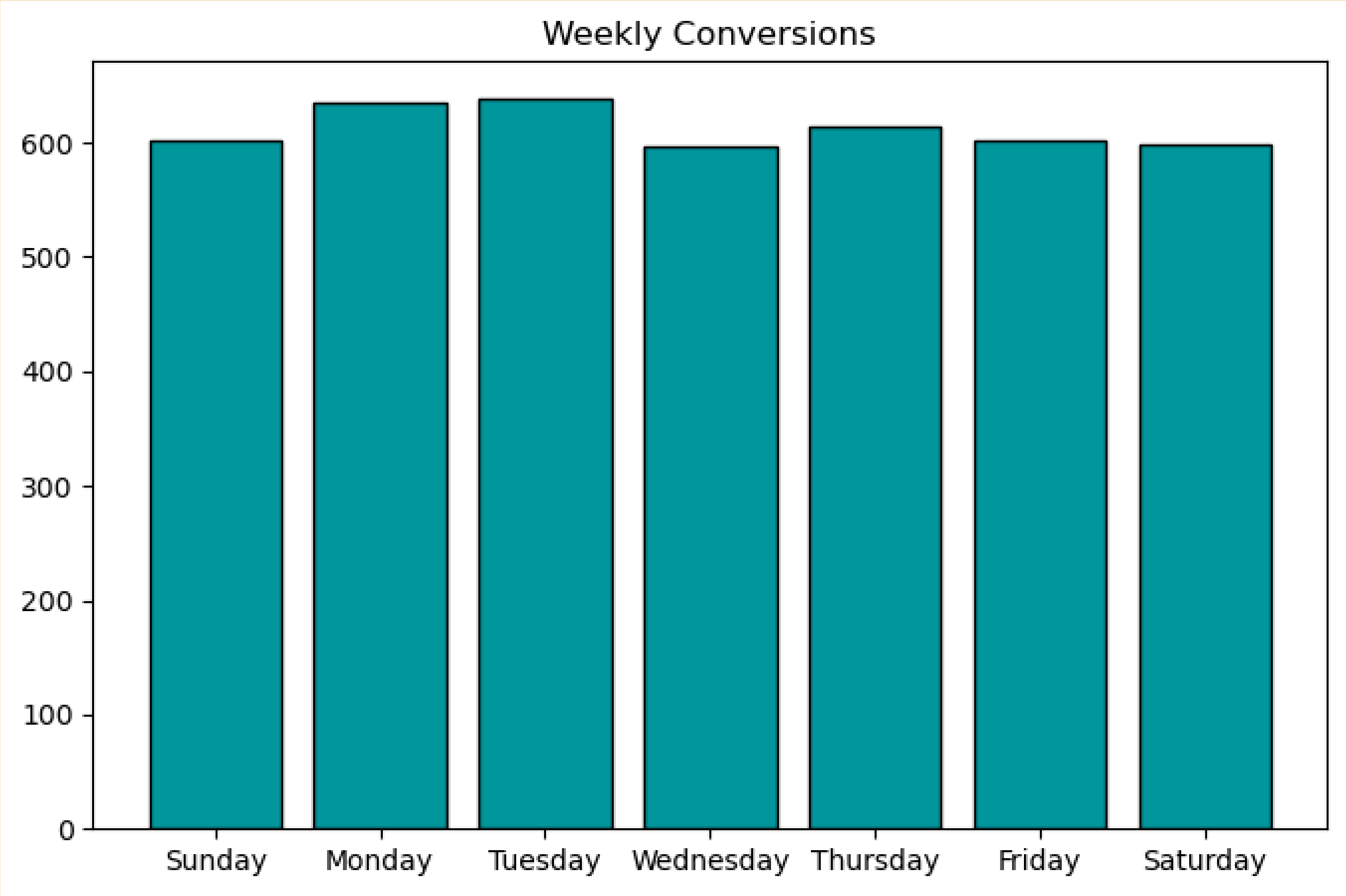
For 50 Clicks, Expected Conversion : 13.0
For 80 Clicks, Expected Conversion : 19.31
```

- Model Accuracy: The Linear Regression model demonstrates strong predictive power with an R^2 score of 76.35%, indicating a reliable correlation between Facebook ad clicks and conversions.
- Strategic Planning: These insights enable data-driven decisions regarding resource allocation, budget planning, and campaign optimization.
- Performance Forecasting: Predicted conversion values (e.g., 13.0 conversions for 50 clicks) allow for realistic goal setting, precise ad spend optimization, and accurate ROI assessment.

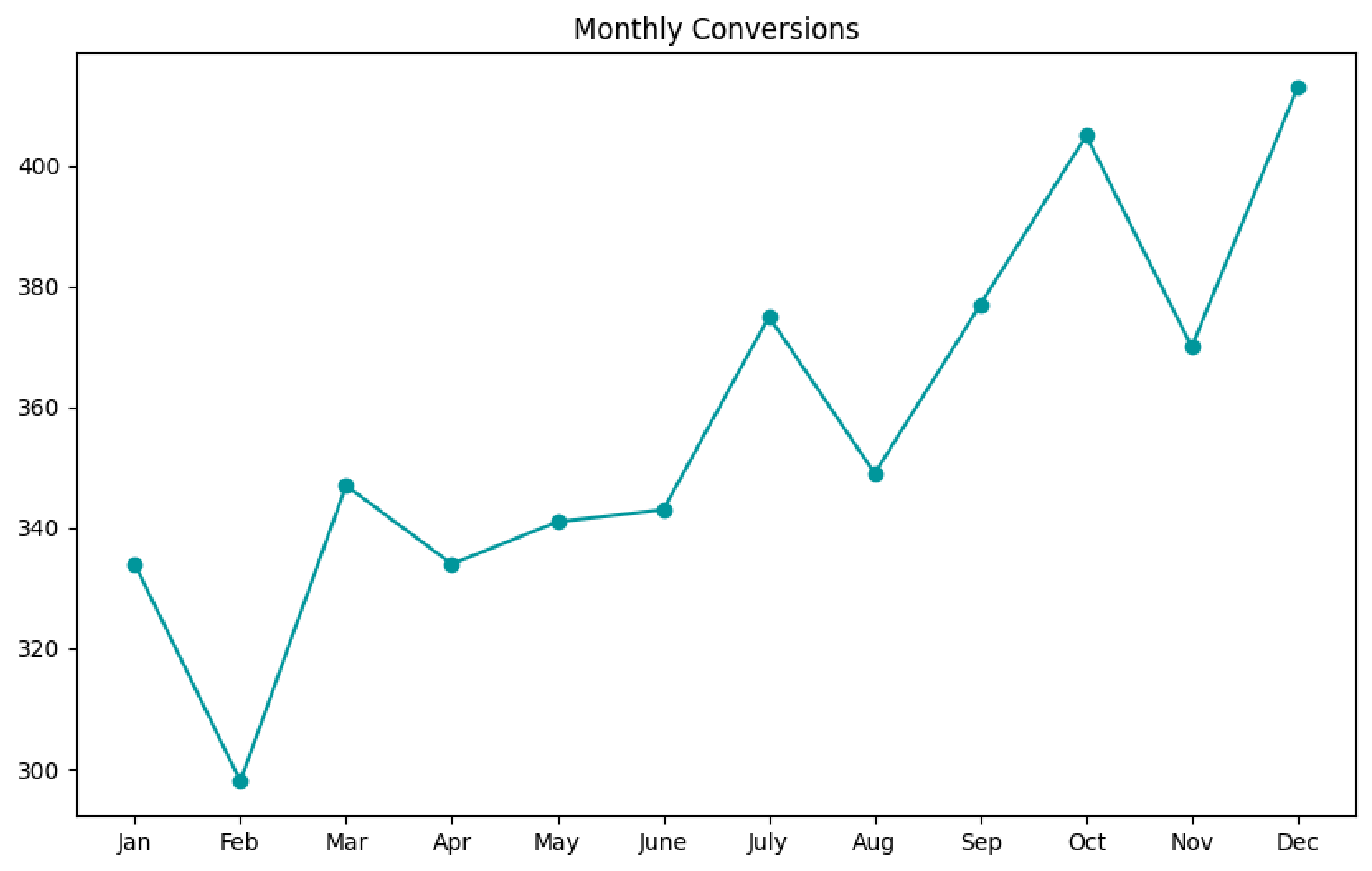
Analysing Facebook Campaign Metrics Over Time

Weekly Trend



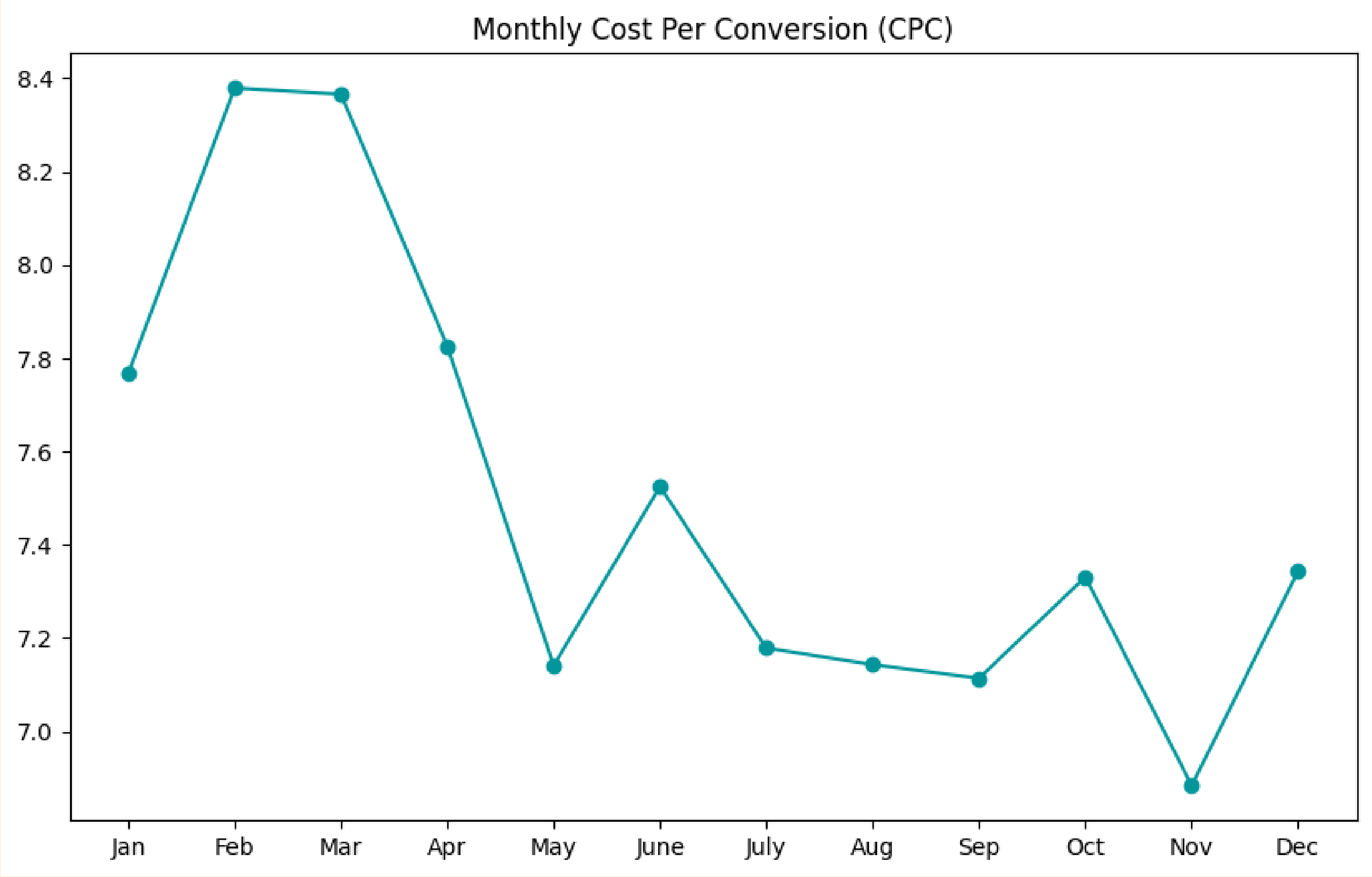
- Early-Week Performance Peak: Conversions hit their highest levels on Monday and Tuesday, with Tuesday serving as the week's peak performer (exceeding 600 units).
- Mid-Week Dip & Recovery: There is a noticeable drop in volume on Wednesday, followed by a moderate recovery on Thursday before stabilizing through the weekend.
- High Stability: Despite daily fluctuations, performance remains highly consistent, maintaining a floor of approximately 600 conversions across all seven days.

Monthly Trend



- Strong Upward Momentum: The overall data shows a significant positive monthly trend, climbing from a low of approximately 300 in February to a peak of over 410 in December.
- Q4 Seasonal Peak: Performance surged in the final quarter, with December hitting the annual maximum, indicating a successful end-of-year conversion strategy.
- Mid-Year Volatility: While the overall trend is bullish, the chart shows recurring fluctuations (dips in April, August, and November) following periods of rapid growth.

Cost per Conversions



- Peak Inefficiency (Q1): CPC reached its highest levels in February and March (~8.38), indicating the most expensive period for acquisitions before a sharp decline in April.
- Optimal Performance (Q4): The lowest CPC of the year occurred in November (~6.88), representing the most cost-efficient month for conversions.
- Volatility Spike (Q2): After a major drop in May, June saw a significant rebound spike to ~7.52, suggesting a temporary decrease in campaign efficiency or increased market competition.
- Stabilized Efficiency (Q3): Costs remained remarkably stable and low from July through September, maintaining a tight range between 7.11 and 7.18.

Relationship between Advertising spend and Conversion rates

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Cointegration test score: -14.755428385103219
P-value: 2.1337375979061323e-26

p-value is less than significance value, Reject the null hypothesis
```

- With the p-value near zero, we reject the null hypothesis, confirming a stable, long-term equilibrium between advertising spend and conversion volume.

- This verified relationship allows for data-driven budget scaling. By prioritizing high-ROI campaigns and adjusting spend based on this predictable trend, we can maximize conversion growth while maintaining cost efficiency.